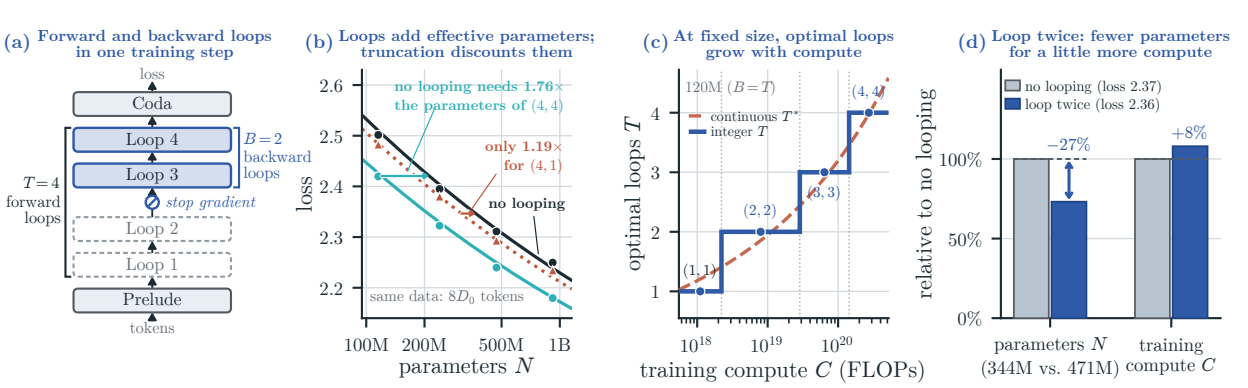
(a) Forward and backward loops in one training step
loss
Coda
Loop 4
Loop 3
Loop 2
Loop 1
Prelude
tokens
B = 2 backward loops
T = 4 forward loops
stop gradient
(b) Loops add effective parameters; truncation discounts them
no looping needs 1.76× the parameters of (4, 4)
only 1.19× for (4, 1)
no looping
same data: 8D0 tokens
loss
parameters N
100M
200M
500M
1B
2.6
2.5
2.4
2.3
2.2
(c) At fixed size, optimal loops grow with compute
120M (B = T)
continuous T*
integer T
optimal loops T
training compute C (FLOPs)
(1, 1)
(2, 2)
(3, 3)
(4, 4)
(d) Loop twice: fewer parameters for a little more compute
no looping (loss 2.37)
loop twice (loss 2.36)
relative to no looping
−27%
+8%
100%
50%
0%
parameters N
(344M vs. 471M)
training compute C